

Multimodal Transformers for Land-Cover Distribution Prediction

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Abstract—Transformer-based architectures have enabled effective multimodal learning by jointly modeling visual and textual information. This study investigates the research question: To what extent can multimodal transformer architectures leverage semantic textual descriptions (without quantitative information) to improve prediction of land-cover class distributions from segmentation masks? The task is formulated as a multi-output regression problem: given a satellite image and its textual caption, predict the percentage composition of seven land-cover classes. Three model types are compared — a Vision Transformer (ViT) image-only baseline, a text-only DistilBERT model, and a multimodal concatenation model combining both encoders — across five caption columns generated by different vision-language models.

Index Terms—multimodal learning, transformers, remote sensing

I. INTRODUCTION

Traditionally, remote sensing land-cover analysis has been based solely on image data. But more recent satellite data sets are offering textual descriptions that are generated by vision-language models because there may be semantic content that is not easily extractable in the raw pixel form of the satellite data. The question of whether these types of descriptions significantly enhance structured predictability of land-cover composition is an open question.

This study addresses the research question: *To what extent can multimodal transformer architectures leverage semantic textual descriptions (without quantitative information) to improve prediction of land-cover class distributions from segmentation masks?* The dataset contains true-color satellite images paired with segmentation masks and five sets of machine-generated captions produced by different models. Ground truth labels are the percentage compositions of seven land-cover classes: tree cover, shrubland, grassland, cropland, built-up area, barren land, and water.

Phase 1 showed proof of feasibility with a proof-of-concept experiment that compared a ViT image-only model with a ViT+DistilBERT concatenation model. Phase 2, which builds upon this, is structured benchmarking study of seven experiments, which addresses two problems identified in Phase 1: over-aggressive leakage filtering and uniform learning rates between pretrained and randomly initialized parameters.

II. LITERATURE SURVEY

Dosovitskiy et al. [1] demonstrated that Vision Transformers (ViT) achieve strong performance on image recognition by

treating image patches as sequential tokens, enabling global dependency modeling without convolutional inductive biases. Radford et al. [2] introduced CLIP, which aligns visual and textual representations through contrastive learning on large-scale image-text pairs, enabling zero-shot cross-modal transfer. Lüddecke and Ecker [3] extended CLIP with CLIPSeg, adding a text-conditioned decoder for image segmentation. Yang et al. [4] proposed LAVT, a language-aware vision transformer that uses cross-attention between visual and linguistic features for referring image segmentation. Xu et al. [5] showed in GroupViT that textual supervision alone can guide visual grouping and segmentation without pixel-level annotations.

The majority of existing multimodal transformer literature targets segmentation, retrieval, or visual grounding tasks. The application of textual supervision to structured regression over land-cover compositions — and specifically the effect of caption quality on prediction — remains understudied. This gap motivates the proposed benchmarking study.

III. METHODS

A. Dataset and Leakage Prevention

The dataset contains satellite images with paired segmentation masks and five caption columns: `vision_qwen3-vl-8b`, `vision_gemma3-4b`, `hybrid_qwen3-vl-8b`, `hybrid_gemma3-4b`, and `text_qwen3-4b`. Vision captions are generated from the image alone; hybrid captions integrate both visual and linguistic model outputs; text captions are generated without image access. Ground truth labels are the seven land-cover percentage compositions derived from segmentation mask pixel counts.

Two forms of data leakage were identified and removed from all caption columns: (1) explicit numeric percentages (e.g., “43%”), which directly encode the regression target; and (2) compositional proportion language (e.g., “predominantly”, “sparse”, “majority”), which qualitatively encodes coverage amounts derived from mask statistics. Both categories were removed using a regular expression filter applied uniformly to all five caption columns before any model training.

A stratified 70/15/15 train/validation/test split was applied using the dominant land-cover class as the stratification key, ensuring balanced class representation across all splits.

B. Model Architectures

Three model architectures are evaluated. The **ViT baseline** encodes each image with ViT-Base/16 [1] pretrained on ImageNet-21k. The [CLS] token representation is passed to a two-layer regression head (Linear–ReLU–Dropout(0.2)–Linear) to predict seven land-cover percentages.

The **multimodal concatenation model** encodes images with the same ViT and captions with DistilBERT-base-uncased [6]. The [CLS] tokens from both encoders are concatenated and passed to a three-layer regression head (Linear–ReLU–Dropout(0.2)–Linear–ReLU–Dropout(0.2)–Linear).

The **text-only model** encodes only the caption with DistilBERT and passes its [CLS] token through a two-layer regression head. This model establishes a lower bound to quantify the contribution of visual features.

C. Training Setup

All models are trained with MSE loss and the Adam optimizer. Differential learning rates are applied: pretrained encoder parameters (ViT, DistilBERT) use $lr = 10^{-5}$ to preserve pretrained representations, while randomly initialized regression head parameters use $lr = 10^{-4}$. Phase 1 proof-of-concept models were trained for 7 epochs; Phase 2 benchmarking models for 10 epochs. All experiments are tracked with Weights & Biases. Evaluation metrics are Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), both in percentage points (%).

IV. EXPERIMENTS AND RESULTS

A. Experiment Design

Seven experiments are run systematically, varying model type and caption column:

- **Exp 0:** Image-only ViT baseline (no text input).
- **Exp 1:** Multimodal with vision-generated captions (safe_vision_qwen3-vl-8b).
- **Exp 2:** Text-only DistilBERT (no image input).
- **Exp 3:** Multimodal with shuffled captions — breaks image-text alignment to test whether improvement is due to genuine semantic correlation.
- **Exp 4:** Multimodal with filtered hybrid Gemma captions.
- **Exp 5:** Multimodal with filtered hybrid Qwen captions.
- **Exp 6:** Multimodal with filtered text-only Qwen captions.

B. Results

Table I reports validation and test MAE and RMSE for all seven experiments. Fig. 1 visualizes test MAE across experiments.

C. Analysis

Caption quality is the decisive factor. Vision-only captions (Exp 1, MAE 2.87) perform worse than the image-only baseline (Exp 0, MAE 2.73). In contrast, hybrid and text-generated captions (Exp 4–6) reduce test MAE to 1.79–1.93, a 29–34% improvement over baseline. This shows that more informative caption sources are complementary in providing semantic

TABLE I
PHASE 2 RESULTS (MAE/RMSE IN %). BEST IN BOLD.

Exp	Model	V-MAE	V-RMSE	T-MAE	T-RMSE
0	ViT	2.67	5.89	2.73	6.05
1	+vision_qwen	2.86	6.15	2.87	6.19
2	Text-only	6.32	13.13	6.34	13.17
3	+shuffled	2.85	6.10	2.91	6.23
4	+hyb_gemma	1.89	3.76	1.93	3.87
5	+hyb_qwen	1.82	3.57	1.83	3.60
6	+text_qwen	1.74	3.33	1.79	3.49

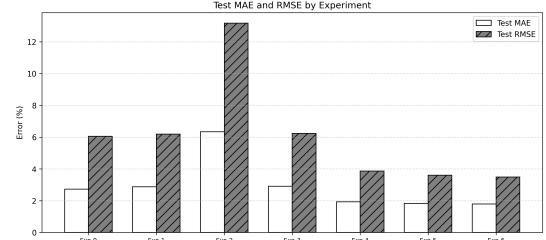


Fig. 1. Test MAE (%) for all seven Phase 2 experiments. Lower is better. Vision-only captions (Exp 1) provide no improvement over the image baseline (Exp 0), while hybrid and text-generated captions (Exp 4–6) reduce MAE by 29–34%.

information that cannot be obtained after leakage filtering by the image alone.

The shuffled caption ablation confirms genuine alignment.

Exp 3 (shuffled captions, MAE 2.91) performs worse than Exp 1 (real vision captions, MAE 2.87) and dramatically worse than Exp 5 and 6. This demonstrates that the multimodal model learns meaningful image-text correspondence rather than exploiting statistical regularities in the text distribution.

Text alone is insufficient. The text-only model (Exp 2, MAE 6.34) is more than twice as bad as the image baseline, confirming that vision is the primary modality and text serves as a complementary signal.

Phase 1 inconsistency resolved. In Phase 1, the multimodal model underperformed the image baseline due to two issues: the leakage filter removed semantically useful proportion words along with true leakage, and uniform learning rates caused gradient interference between pretrained and randomly initialized parameters. Both issues were corrected in Phase 2, yielding consistent and interpretable results.

V. CONCLUSION

This study demonstrates that multimodal transformers can substantially improve land-cover composition prediction in remote sensing when caption quality is sufficient. Vision-only captions provide no improvement over the image baseline, while hybrid and text-generated captions — which integrate richer semantic content — reduce MAE by up to 34%. The shuffled caption ablation confirms that these gains reflect genuine cross-modal semantic alignment. Future work in Phase 3 will investigate architectural variations including cross-attention fusion and frozen vs. fine-tuned text encoders to further understand how and where textual information contributes to prediction accuracy.

VI. REPRODUCIBILITY

GitHub Repository:

<https://github.com/ilaydaergur/multimodal-landcover-transformers>

Weights & Biases Report:

W&B Experiment Report

REFERENCES

- [1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. 9th Int. Conf. Learn. Representations (ICLR)*, Virtual, May 2021.
- [2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, pp. 8748–8763, Jul. 2021.
- [3] T. Lüddecke and A. Ecker, “Image segmentation using text and image prompts,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, pp. 7086–7096, Jun. 2022.
- [4] Z. Yang, J. Wang, J. Tang, K. Chen, H. Zhao, and P. H. S. Torr, “LAVT: Language-aware vision transformer for referring image segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, pp. 18818–18828, Jun. 2022.
- [5] J. Xu, S. De Mello, S. Liu, W. Byeon, T. Breuel, J. Kautz, and X. Wang, “GroupViT: Semantic segmentation emerges from text supervision,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, pp. 18134–18144, Jun. 2022.
- [6] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” in *Proc. 5th Workshop Energy Efficient Mach. Learn. Cogn. Comput. (EMC2-NIPS)*, Vancouver, Canada, Dec. 2019.